Infinite Series 4

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Problem 1

The sum can be written as a binary expansion:

$$\sum_{p \text{ prime}} \frac{1}{2^p} = 0.d_1 d_2 d_3 \ldots,$$

$$d_n = \begin{cases} 1 & n \text{ prime} \\ 0 & \text{otherwise.} \end{cases}$$

The sum is rational if the decimal expansion repeats. Assuming it does, consider a repeating block of length k:

$$(d_N, d_{N+1}, \ldots, d_{N+k-1}).$$

This block cannot be all 0s (since that would mean there are finitely many primes), so there is at least one 1, and the number of consecutive zeros between two 1s is bounded by some $B < k$. This would imply that the gaps between primes are eventually bounded, which is false, so the sum is irrational.

Problem 2

Let $x = \frac{\arccos(1/3)}{\pi}$ and assume x is rational. So

$$\frac{\arccos(1/3)}{\pi} = x, \quad \cos(x\pi) = \frac{1}{3}.$$

The only rational multiples of π that have a rational cosine are:

$$\begin{aligned} \cos(0) &= 1, \\ \cos(\pi/3) &= 1/2 \\ \cos(\pi/2) &= 0, \\ \cos(2\pi/3) &= -1/2 \\ \cos(\pi) &= -1, \end{aligned}$$

none of which are $1/3$. So x must be irrational.

Problem 3

Use the binomial theorem:

$$x^n(1-x)^n = \sum_{k=0}^{n} \binom{n}{k}(-1)^k x^{n+k}.$$

Differentiate n times:

$$\frac{d^n}{dx^n}x^{n+k} = \frac{(n+k)!}{k!}x^k,$$

$$\frac{d^n}{dx^n}(x^n(1-x)^n) = \sum_{k=0}^{n} \binom{n}{k}(-1)^k \frac{(n+k)!}{k!}x^k.$$

Divide by $n!$:

$$P_n(x) = \sum_{k=0}^{n} \binom{n}{k}(-1)^k \frac{(n+k)!}{n!k!}x^k = \sum_{k=0}^{n}(-1)^k \binom{n}{k}\binom{n+k}{k}x^k.$$

Since $(-1)^k\binom{n}{k}\binom{n+k}{k}$ is an integer, every coefficient of $P_n(x)$ is an integer.

Problem 8

Fix irrational α which is a root of an n-degree polynomial with integer coefficients. For any integers p, q with p/q sufficiently close to α, $|f(p/q)| \geq \frac{1}{q^n}$, because each term is an integer divided by q^n, and the result is not 0.

Now factor $|f(r)|$ for any rational r:

$$|f(r)| = |a_n| \prod_{i=1}^{n} |r - \alpha_i| = |a_n||r-\alpha| \prod_{i=2}^{n} |r - \alpha_i|.$$

Our goal is to bound the product over $i \geq 2$. Fix

$$B = 1 + \max\{|\alpha|, |\alpha_2|, \ldots, |\alpha_n|\}.$$

Assuming $r = p/q$ is within 1 of α (because otherwise the inequality holds for any $A \leq 1$), we have

$$|r - \alpha| < 1 \implies |r| \leq |\alpha| + 1 \leq B.$$

Now we can bound each factor $|r - \alpha_i|$:

$$|r - \alpha_i| \leq |r| + |\alpha_i| \leq B + \max_{2 \leq i \leq n} |\alpha_i|.$$

Let

$$C = \prod_{i=2}^{n} (B + \max_{2 \leq i \leq n} |\alpha_i|).$$

Then for all r with $|r - \alpha| < 1$,

$$|f(r)| \leq |a_n|C|r - \alpha|.$$

So we have that

$$\frac{1}{q^n} \leq |f(r)| \leq |a_n|C|r - \alpha|,$$

$$|r - \alpha| \geq \frac{1}{|a_n|Cq^n}.$$

So if we let $A = \frac{1}{|a_n|C}$, the inequality holds.

Problem 9

For contradiction, assume α is algebraic and irrational. By Liouville's theorem, there exists $A > 0$ such that, for all p, q,

$$\left|a - \frac{p}{q}\right| > \frac{A}{q^d},$$

where d is the degree of the polynomial of which α is a root. Now, choose any integer $n > d$. For large enough q, $1/q^n < A/q^d$. For those large q, the inequalities

$$\left|\alpha - \frac{p}{q}\right| > \frac{A}{q^d}, \quad \left|\alpha - \frac{p}{q}\right| < \frac{1}{q^n}$$

contradict each other. Therefore, α is not algebraic, so it's transcendental.